# AUTOMATED WILDLIFE MONITORING AND DETERRENT SYSTEM

*PROJECT REPORT PHASE- II*

*submitted by*

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*in partial fulfilment of the requirements*
*for the award of the degree of*

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**DEPARTMENT OF ELECTRONICS ENGINEERING**
**SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)**
**KOTTAYAM**

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# CERTIFICATE

This is to certify that this report entitled **AUTOMATED WILDLIFE MONITORING AND DETERRENT SYSTEM** is the bonafide record Major Project Report submitted by **MIDHUN MATHEW (MGP21UEC036)** during the academic year **2024-2025** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in **Electronics and Communication Engineering** of the Saintgits College of Engineering (Autonomous) under the APJ Abdul Kalam Technological University.

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# ABSTRACT


**KEYWORDS :-** ***Wildlife Monitoring*****, Deep Learning, LoRa Communication, YOLO, Human-Wildlife Conflict, Crop-Protection**

Human-wildlife conflicts, particularly those that involve elephants, pose a significant threat to agricultural communities and forest edge settlements. To address this issue, an Automated Wildlife Monitoring and Deterrent System has been developed. The system leverages deep learning-based object detection models, specifically an optimized YOLOv11n network, to identify wildlife such as elephants, tigers, monkeys, squirrels, and bats from live camera feeds. Upon detecting an animal, the system initiates a multilayered deterrence mechanism using a carbide-based acetylene gas generator that produces loud sounds and bright flashes to repel wildlife without causing harm. Real-time alerts and system statuses are communicated to nearby residents and forest officials using LoRa wireless modules, ensuring timely intervention even in remote regions with limited network connectivity. The entire setup is solar powered, enabling sustainable and continuous operation. In addition, sensor modules ensure the safe and efficient functioning of the acetylene generation mechanism. Compared to traditional methods such as electric fencing and manual patrolling, the proposed system offers a more intelligent, eco-friendly, and automated solution to mitigate human-wildlife conflicts.

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# CHAPTER 1

# INTRODUCTION

Human-wildlife conflicts, particularly those involving elephants and other large mammals, have emerged as a serious concern for rural and forest-adjacent communities. Traditional methods for preventing wildlife intrusions, such as manual patrolling, electric fencing, and the use of scare devices, often suffer from limited effectiveness, high maintenance costs, and potential harm to both humans and animals. These methods can also be labour-intensive, unreliable under varying environmental conditions, and dependent on continuous human intervention. To address these challenges, there is a growing need for intelligent, automated, and sustainable systems capable of detecting wildlife intrusions in real-time and triggering effective non-lethal deterrent actions.

Recent advances in deep learning, particularly in real-time object detection models like YOLOv11n, have opened new possibilities for automated wildlife monitoring. By deploying edge-optimized AI models capable of accurately identifying various animal species from live video feeds, it is possible to create proactive and responsive monitoring systems that do not rely on manual surveillance.

Wireless communication technologies, such as LoRa, offer robust long-range data transmission with minimal power consumption, making them ideal for deployment in remote forest-edge regions. Integrating these technologies with intelligent wildlife detection systems enables real-time alerts and activation of deterrent mechanisms, even in areas with poor connectivity to the cellular network.

This project aims to develop a novel method for mitigating human-wildlife conflict through an Automated Wildlife Monitoring and Deterrent System a carbide-based deterrent mechanism that produces loud sounds and flashes to safely repel the animals.

This approach has the potential to revolutionize conflict mitigation strategies, providing an eco-friendly, low-cost, and scalable solution to protect farmland and human settlements from wildlife intrusions. By promoting nonlethal deterrence and real-time alerting, this system supports both conservation efforts and community safety, fostering sustainable coexistence between humans and wildlife.

# CHAPTER 2

# LITERATURE SURVEY

## 2.1 IoT-Based Animal Trespass Identification and Prevention System for Smart Agriculture [1]

**Manoharan S., Vignesh M., Ranjith R. "IoT-Based Animal Trespass Identification and Prevention System for Smart Agriculture."** *International Research Journal of Engineering and Technology (IRJET)* 7.5 (2020): 2395-0056.

Wildlife intrusion into agricultural lands causes significant damage to crops and property. In this paper, an IoT-based system is proposed to prevent animal intrusions using motion sensors, thermal imaging, and AI-based classification. While the system provides real-time alerts to farmers, a major limitation is its dependency on external network connectivity and continuous power supply, which restricts deployment in remote areas. Our project addresses these challenges by employing a solar-powered, low-power wide-area communication (LoRa) based architecture, ensuring uninterrupted operation in off-grid environments with minimal energy requirements.

## 2.2 Wild Animals Intrusion Detection for Safe Commuting in Forest Corridors Using AI Techniques [2]

**Ashwini J., Anandhi S. "Wild Animals Intrusion Detection for Safe Commuting in Forest Corridors Using AI Techniques."** *International Journal of Engineering Research Technology (IJERT)*, vol. 10, no. 11, 2021.

This study proposes the use of AI-based image recognition models to detect wild animals in forest corridors, aimed at preventing accidents. However, reliance on pre-installed fixed surveillance cameras limits the system's coverage and real-time effectiveness due to environmental obstructions. In contrast, our system incorporates mobile and strategically placed camera modules combined with real-time edge computing, enabling wider surveillance with higher adaptability even in dynamically changing forest and farmland environments.

## 2.3 IoT-Based Smart Farmland Using Deep Learning [3]

**Neha K. Chaurasia, Sushma B. "IoT-Based Smart Farmland Using Deep Learning."** *International Journal of Engineering Research Technology (IJERT)*, vol. 9, no. 6, 2020.

An IoT-based security solution for farmlands using deep learning is proposed in this study, employing CNNs for animal detection and activating static sound deterrents. However, static deterrents tend to lose effectiveness over time due to animal habituation. Our project overcomes this by implementing a dynamic, carbide-based acetylene deterrent mechanism that generates unpredictable loud sounds and flashes, reducing the likelihood of animals adapting and ensuring sustained deterrent effectiveness.

## 2.4 Monitoring the Movements of Wild Animals and Alert System Using Deep Learning Algorithm [4]

**K. Anusha, P. Gnanaprakasam, A. Kalaiselvi "Monitoring the Movements of Wild Animals and Alert System Using Deep Learning Algorithm."** *International Journal of Research in Engineering, Science and Management (IJRESM)*, vol. 4, no. 8, 2021.

This paper discusses real-time monitoring systems utilizing deep learning algorithms for animal detection. However, cloud-based AI processing introduces latency and necessitates constant network availability, limiting its application in rural settings. Our project circumvents these limitations by deploying lightweight, edge-optimized AI models such as YOLOv11n directly on embedded devices like Raspberry Pi, enabling offline processing with minimal latency and immediate activation of deterrent mechanisms without dependency on cloud servers.

## 2.5 Development of an Elephant Detection and Repellent System Based on EfficientDet-Lite Models [5]

**P. Prakash, R. Ashok Kumar, V. Sriram, S. Srikanth "Development of an Elephant Detection and Repellent System Based on EfficientDet-Lite Models."** *International Research Journal of Engineering and Technology (IRJET)*, vol. 8, no. 6, 2021.

In this study, an EfficientDet-Lite-based model was utilized for elephant detection, coupled with pre-recorded sound-based deterrence. While effective initially, static audio deterrents may lead to animal desensitization over time. Our project proposes a more robust solution by combining real-time wildlife detection with a physical deterrent mechanism involving controlled acetylene gas ignition to produce loud acoustic and visual stimuli. This method increases unpredictability, enhancing long-term deterrence without harming the animals.

# CHAPTER 3

# OBJECTIVES

## 3.1 Development of an Automated Wildlife Monitoring System

The project aims to design and develop a fully automated wildlife monitoring system capable of real-time detection and classification of wild animals, particularly elephants, using optimized deep learning models. By leveraging computer vision techniques and wide-angle surveillance cameras, the system ensures continuous observation of vulnerable agricultural and forest-border regions.

## 3.2 Implementation of an Eco-Friendly Deterrent Mechanism

A key objective of the system is to implement a non-lethal, reusable deterrent mechanism utilizing a carbide-based acetylene gas generation process. Upon detection of wildlife intrusions, the system automatically activates the deterrent, producing loud sounds and bright flashes to scare away animals without inflicting harm. This approach promotes humane wildlife management practices and reduces the need for traditional hazardous methods like electric fencing or chemical repellents.

## 3.3 Integration of Long-Range, Low-Power Communication

To ensure real-time alerting and coordination, the project integrates LoRa-based wireless communication technology. This long-range, low-power system enables efficient transmission of alerts and system status updates across several kilometers, even in remote areas with limited or no mobile network coverage. The use of LoRa ensures timely notification to farmers, residents, and forest officials, facilitating prompt human intervention when necessary.

## 3.4 Deployment of Sustainable, Solar-Powered Infrastructure

Recognizing the challenges of deploying electronic systems in remote and forested areas, the project prioritizes the use of solar-powered energy solutions. The system is designed to operate autonomously with minimal maintenance, utilizing solar panels and battery backup to ensure uninterrupted functioning. Energy-efficient hardware design and low-power operation modes further contribute to sustainability, enabling continuous 24/7 operation even during adverse weather conditions.

## 3.5 Real-Time System Monitoring and Maintenance Automation

The system incorporates sensor modules such as HX711 weight sensors and water level detectors to monitor the critical resources required for acetylene production. Automated monitoring ensures timely refilling of materials and prevents operational downtime. Alerts related to low resource levels are automatically communicated to the user, minimizing the need for manual inspection and reducing system maintenance efforts.

## 3.6 Field Validation and Performance Optimization

An essential objective is the rigorous field testing and validation of the system across multiple agricultural zones prone to wildlife intrusions. Performance metrics such as detection accuracy, latency, wireless communication reliability, and energy consumption are systematically evaluated to refine the system for real-world deployment. The goal is to achieve high detection precision, minimal response delays, and maximum deterrent effectiveness under varied environmental conditions.

## 3.7 Contribution to Human-Wildlife Conflict Mitigation

The system not only protects human lives and agricultural resources but also promotes wildlife conservation by using humane deterrence methods. Long-term, it seeks to foster peaceful coexistence between human settlements and wildlife habitats.

# CHAPTER 4

# METHODOLOGY

The proposed system is designed for real-time detection and deterrence of wild animals in forest-edge or agricultural areas. It integrates edge-based AI, long-range wireless communication, and automated deterrent mechanisms powered sustainably through solar energy. The methodology is structured around a two-stage architecture comprising a Transmitter Unit and a Receiver Unit, each responsible for a defined sequence of operations.

## 4.1 Block Diagram

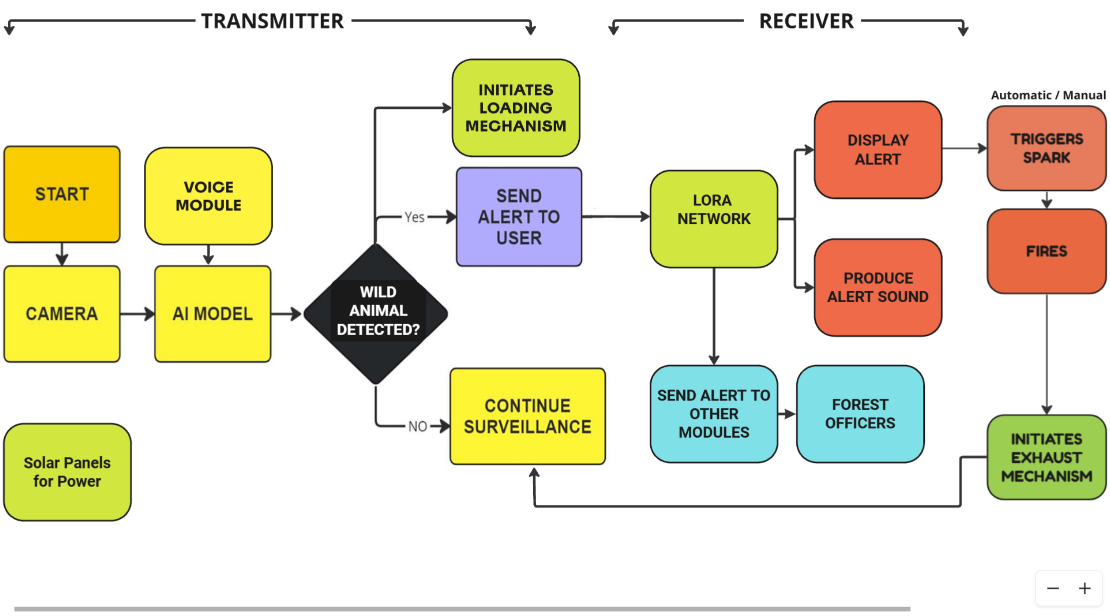



Figure 4.1: Block Diagram

### 4.1.1 Transmitter Unit

The transmitter unit forms the core of the detection system. It begins operation when powered on via solar panels connected to a 12V battery and managed using a Battery

Management System (BMS) to ensure uninterrupted performance in outdoor environments. Upon initiation, a camera module connected to a Raspberry Pi 4B captures real-time images or video. These inputs are immediately processed by a YOLO-based deep learning model deployed locally on the Raspberry Pi. This lightweight convolutional neural network has been trained on a custom dataset with five wildlife classes: elephant, tiger, monkey, squirrel, and bats. The use of edge-based inference reduces latency and eliminates the need for internet connectivity, a crucial feature for remote environments.

To complement visual detection, a voice module embedded within the transmitter can issue audible warnings when the system is activated, both to deter animals and alert nearby humans. Once an animal is detected, a control logic block evaluates the model's output. If a wild animal is identified, the system initiates multiple parallel operations. First, it triggers a mechanism that begins preparing the deterrent response. This includes activating a peristaltic pump to fill a carbide chamber with water, allowing acetylene gas to be produced for later ignition. Simultaneously, an alert is sent via the LoRa (Long Range) wireless communication module to the receiver unit. The LoRa SX1278 module is selected for its ability to transmit low-power RF signals over several kilometres, making it ideal for covering vast agricultural or forest boundaries. If no animal is detected, the system simply continues to monitor the environment in surveillance mode, thereby conserving energy and system resources.

### 4.1.2 Receiver Unit

The receiver unit is responsible for alert dissemination, deterrent activation, and further system escalation. Upon receiving a signal from the transmitter via LoRa, the ESP32 microcontroller processes the data and triggers a series of alerts. A buzzer module generates a loud warning sound, and a connected LCD display provides visual status information, informing users or nearby personnel of the threat. The system supports both automatic and manual operation modes, allowing users to intervene when necessary.

In automatic mode, the receiver triggers a high-voltage spark mechanism to ignite the acetylene gas mixture that was earlier prepared. This ignition produces a loud blast intended to startle and repel animals from the protected area. Following this action, the system activates an exhaust mechanism to safely disperse any remaining gases from the firing chamber, ensuring it is safe and ready for the next cycle.

In parallel, the receiver system can forward the alert to other critical endpoints, such as mobile devices used by forest officers or secondary deterrent modules placed in nearby zones. This multi-node alert system ensures that authorities are promptly informed and that wildlife deterrence is applied over a wider area, enhancing the overall effectiveness of the solution..

### 4.1.3 Data Acquisition and Analysis

Capturing movement data and images from the sensors and analyzing them locally using lightweight machine learning models. This step ensures quick, efficient identification of wildlife presence without relying heavily on cloud computation.

### 4.1.4 Wireless Communication

Using LoRa technology to transmit detection alerts over long distances to a centralized monitoring system or directly to deterrent devices. This provides reliable communication even in remote or difficult terrains.

### 4.1.5 Machine Learning Model Development

Training machine learning models to classify detected movement patterns and images as wildlife or non-wildlife entities. Utilizing supervised learning with labeled data to improve model performance and reduce false positives.

### 4.1.6 Deterrent Activation

Upon positive detection of a wild animal, the system activates a **carbide-based deterrence mechanism** designed to produce a loud acoustic burst to scare away intruding animals. This method leverages the chemical reaction between *calcium carbide* ($CaC_2$) and water to generate *acetylene gas* ($C_2H_2$). In the transmitter unit, a PVC chamber is used as a reaction vessel where a measured amount of calcium carbide is placed. Once detection is confirmed, a DC-powered peristaltic pump dispenses a controlled volume of water into this chamber, initiating the reaction:

$$\mathrm{CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2} \tag{4.1}$$

The resulting acetylene gas accumulates within the sealed chamber until a spark, generated via a high-voltage ignition module, ignites the gas-air mixture. This ignition produces a sudden and loud combustion sound, which serves as an effective deterrent to large animals such as elephants and wild boars.

### 4.1.7 Integrated Power and Energy Efficiency

To ensure uninterrupted performance in remote locations, the entire system is powered using solar energy. A 12V solar panel charges a 20,000 mAh lithium-ion battery, which powers both the Raspberry Pi and ESP32 microcontrollers. A smart Battery Management System (BMS) regulates charging, prevents over-discharge, and supports system longevity. This design eliminates dependence on external power infrastructure and provides resilience against environmental challenges.

### 4.1.8 Advantages Over Traditional Approaches

Unlike manual patrolling or fixed noise deterrents, the proposed methodology offers intelligent, automated detection and deterrence with minimal human input. It operates continuously, adapts to real-time wildlife presence, and scales easily across large terrain. The use of solar power, LoRa-based communication, and embedded AI processing ensures that the system is not only sustainable but also cost-effective, accurate, and reliable. Its modular design allows further expansion or replication across critical zones prone to wildlife conflict.

# CHAPTER 5

# SYSTEM DESCRIPTION

The system integrates modern embedded hardware and intelligent software to detect the presence of wildlife and activate deterrent mechanisms in real-time. Hardware components include a camera module, a Raspberry Pi processing unit, LoRa communication modules, a solar-powered energy setup, and a carbide-based deterrent mechanism. Software components such as an optimized deep learning model (YOLOv11n), real-time video processing tools (OpenCV), and wireless communication protocols enable efficient animal detection, alert generation, and deterrent activation.

## 5.1 Hardware Specifications

### 5.1.1 Camera Module (USB Webcam)

Captures real-time video feeds from the monitored environment with high-resolution and wide-angle capabilities. The camera acts as the primary sensory input, providing the necessary image data for wildlife detection, even in low-light conditions with infrared capability if needed.


Figure 5.1: Camera Module

### 5.1.2 Raspberry Pi 4B

Serves as the main processing unit responsible for running the deep learning inference model (YOLOv11n) and controlling the overall system operation. It manages real-time image processing, decision making, communication handling, and deterrent activation processes.

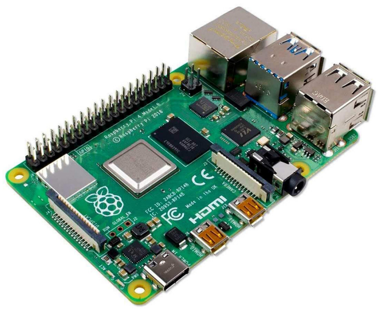


Figure 5.2: Raspberry Pi 4B

### 5.1.3 ESP32 Microcontroller

Acts as the central control unit responsible for coordinating all hardware components, including sensors, actuators, display, and communication modules. With built-in Wi-Fi, Bluetooth, and multiple GPIOs, the ESP32 efficiently manages data acquisition, LoRa communication, and output control. Its high processing speed and low power consumption make it ideal for real-time embedded applications in field environments.

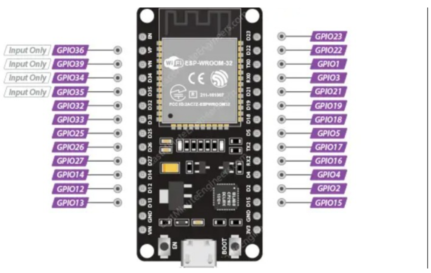



Figure 5.3: ESP32 Microcontroller

### 5.1.4 LoRa SX1278 Module

Ensures long-range, low-power wireless communication between the transmitter (detection unit) and the receiver (deterrent unit). It allows the transmission of alerts and system status messages over several kilometers, making it ideal for remote and forested areas with poor cellular connectivity.

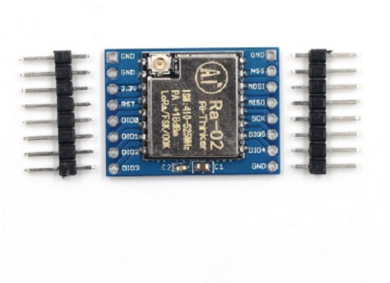


Figure 5.4: LoRa SX1278 Module

### 5.1.5 Solar Panel and Battery Backup System

Provides continuous, renewable energy to power the system components. The solar panel charges a high-capacity lithium-ion battery that ensures the system's uninterrupted operation during nighttime or cloudy weather, promoting off-grid sustainability.

Figure 5.5: Solar Panel and Battery Backup System

### 5.1.6 TFT Display Module

Provides a user-friendly graphical interface for visualizing detection results, system status, and real-time alerts. With its high-resolution display and wide viewing angle, it enables effective monitoring even in outdoor conditions. The display, driven by the ESP32 using the LVGL graphics library, supports intuitive UI layouts for field deployment.

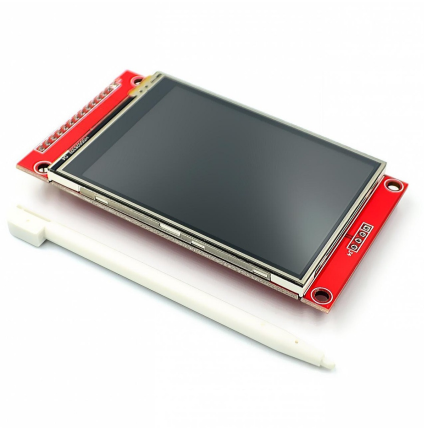


Figure 5.6: 2.8-inch TFT Display Module

### 5.1.7 Solenoid Valve

Controls the release of pressurized water or gas in response to detected wildlife presence. It is activated by the ESP32 to serve as part of the deterrent mechanism, creating sudden bursts of fluid to scare away animals. The valve operates reliably on a 24V power supply, offering fast switching, low power draw, and rugged performance suitable for outdoor conditions.

Figure 5.7: 24V Solenoid Valve

### 5.1.8 Relay Module

Acts as a crucial control interface between the ESP32 and high-voltage components such as the plasma ignition system and the peristaltic pump. It allows the microcontroller to safely switch these high-power devices by providing electrical isolation and precise activation timing. The relay ensures reliable operation of the ignition mechanism and water pumping process essential for acetylene gas generation and controlled ignition.

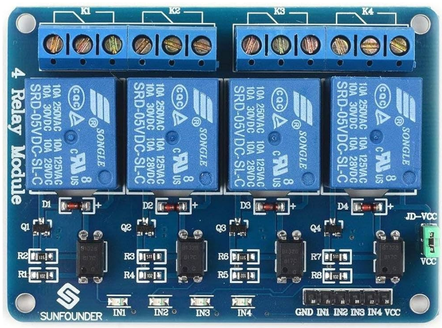



Figure 5.8: Relay Module

### 5.1.9 5V Peristaltic Pump

Used to deliver controlled amounts of water into the reaction chamber for carbide-based gas generation. The pump ensures safe and precise fluid transfer without contamination, thanks to its tubing-based mechanism. Operating at 5V, it is energy-efficient and ideal for low-flow applications, making it suitable for field deployments where accuracy and reliability are essential.

Figure 5.9: 5V Peristaltic Pump

### 5.1.10 Carbide-Based Deterrent Mechanism

Uses a chemical reaction between calcium carbide and water to produce acetylene gas, which is ignited to generate loud explosion-like sounds and bright flashes. This non-lethal, eco-friendly method effectively deters wildlife without causing harm to animals or the environment.


Figure 5.10: Calcium Carbide

### 5.1.11 Sensors

Monitor the availability of water and acetylene production in the deterrent module. These sensors automate maintenance alerts, ensuring consistent system operation and preventing failure due to resource depletion.

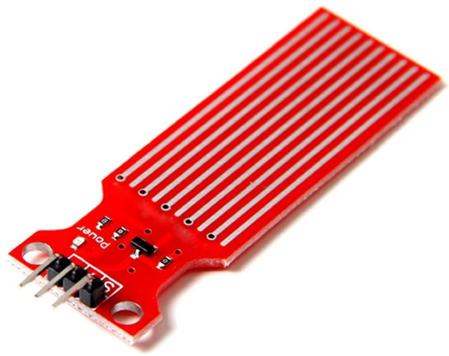


Figure 5.11: Water Level Sensor

## 5.2 Software Specifications

### 5.2.1 Deep Learning Model (YOLOv11n)

The system utilizes a customized YOLOv11n object detection model, optimized for real-time animal detection on resource-constrained hardware. The model is trained on a dataset comprising images of elephants and other wildlife, achieving high detection accuracy while maintaining low computational overhead. Inference is performed using the NCNN framework to maximize efficiency on the Raspberry Pi 4B.

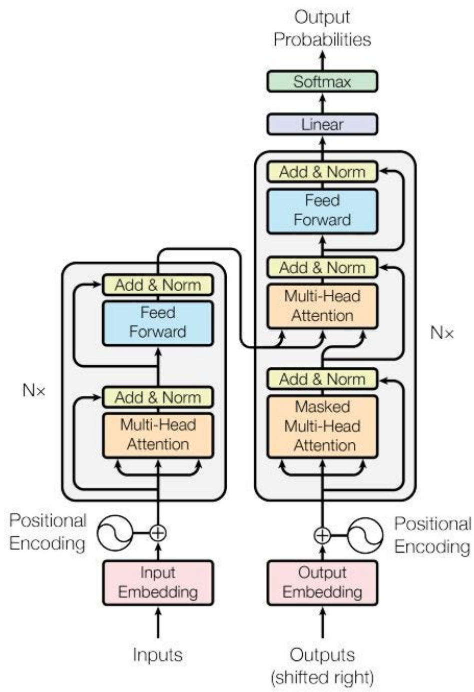



Figure 5.12: YOLO v11 Architecture

### 5.2.2 Real-Time Image Processing (OpenCV)

OpenCV libraries are employed for real-time video frame capture, preprocessing, and handling image data streams. The captured frames are resized, normalized, and fed into the YOLOv11n model for object detection. OpenCV also facilitates annotation and visualization of detection results on the output frames.

### 5.2.3 Wireless Communication Stack (LoRa)

LoRa communication libraries are used to transmit wildlife detection alerts and system status messages from the transmitter (camera and processing unit) to the receiver (deterrent module) and to the user alerting systems. The system ensures robust and reliable communication over long distances, critical for real-world deployment in forested and remote areas.

### 5.2.4 Alert Generation and Deterrent Activation Logic

Upon successful detection of wildlife, the system triggers an alert transmission via LoRa. At the receiver end, a microcontroller (ESP32 or similar) processes the received alert and activates the deterrent system. Visual and acoustic deterrents are triggered sequentially to ensure maximum effectiveness. The activation logic includes safety checks, timeout management, and fallback handling to prevent accidental triggering.

### 5.2.5 Power Management and Energy Optimization

Software modules monitor the energy status using voltage sensing techniques. The Raspberry Pi and LoRa modules are configured with sleep and wake-up cycles to minimize energy consumption during idle periods. Battery levels are monitored, and appropriate alerts are sent if critical low battery levels are detected.

### 5.2.6 Circuit Diagram - Transmitter (Detection Unit)

A lightweight deep learning model (LoRa11n) runs on the Pi to process live video feeds and perform real-time inference. Upon detecting a target animal, the unit sends a wireless alert signal using the LoRa SX1278 module, ensuring long-range and low-power communication to the receiver side. This unit is powered by a regulated power supply and includes an ESP32 microcontroller for coordination and fallback communication.

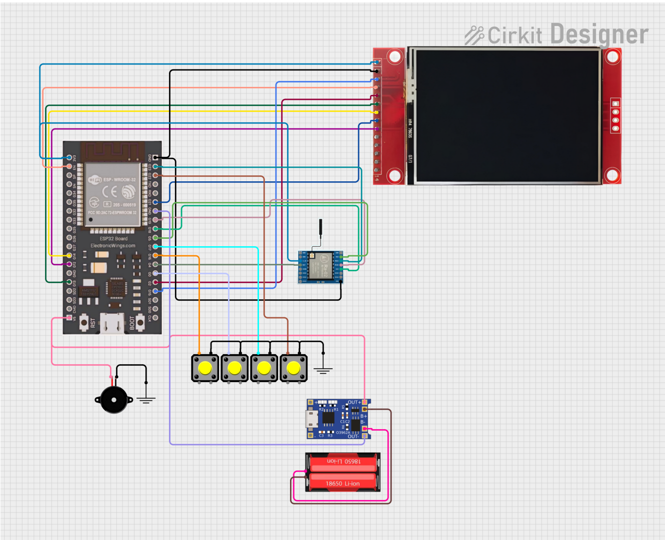



Figure 5.13: Circuit Diagram - Transmitter

### 5.2.7 Receiver Side (Deterrent Activation Unit)

The receiver unit is strategically positioned at a remote deterrent system and is equipped with an Arduino Nano and LoRa SX1278 module. Upon receiving a signal from the transmitter, the Arduino triggers a sequence of deterrent mechanisms. These include activating a 5V peristaltic pump to inject water into a reaction chamber, followed by a solenoid valve that releases carbide to produce acetylene gas. A relay module controls the ignition system (e.g., a plasma arc igniter), creating a loud sound or flash to drive animals away. This unit operates autonomously and reacts instantly to incoming alerts.

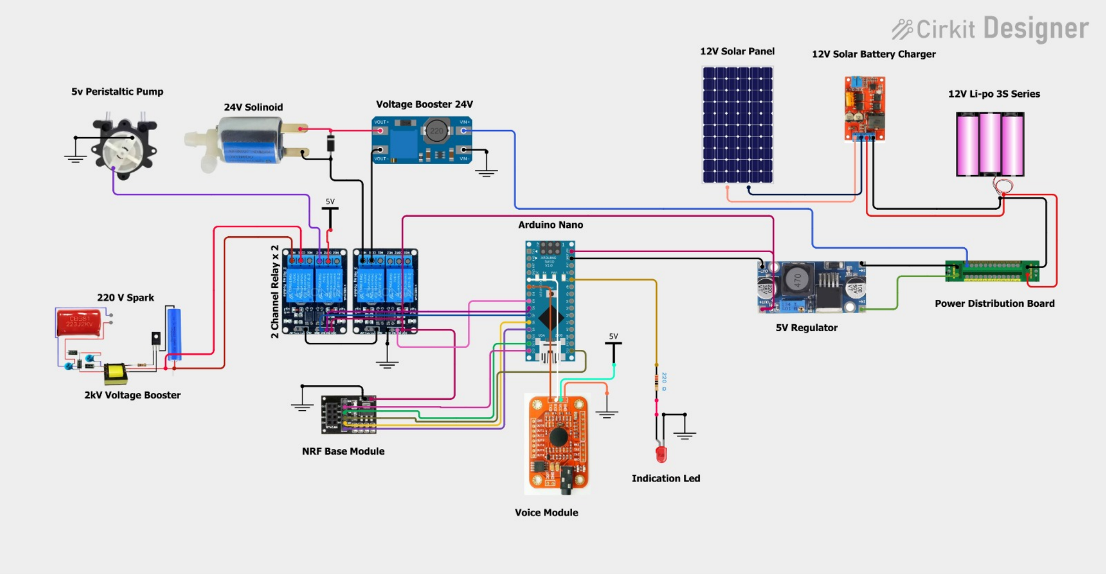



Figure 5.14: Circuit Diagram - Receiver

### 5.2.8 Mechanical Structure and Assembly

The mechanical assembly forms the physical foundation of the deterrent unit, integrating all key components into a functional structure. At the core lies a robust metallic reaction chamber designed to handle gas buildup and ignition pressure safely. A 5V peristaltic pump is connected to a water reservoir to inject precise amounts of water into the carbide holding area. A 24V solenoid valve is positioned beneath the carbide compartment to release acetylene gas into the ignition chamber upon activation. Supporting structures include metal piping for fluid and gas transport, mounting plates for the ESP32 controller and relay modules, and protective casings to shield electronics from weather conditions. The ignition system—powered through a relay—is housed within a heat-resistant casing near the output nozzle. The entire setup is mounted on a stable platform or frame, which can be placed near vulnerable farm boundaries or forest edges for effective deterrence.

Figure 5.15: Prototype - Outdoor Version

Figure 5.16: Prototype - Portable Version

# CHAPTER 6

# SOFTWARE DEVELOPMENT

## 6.1 Dataset Collection

### 6.1.1 Samples

This project involved collecting a diverse set of wildlife images to train and validate the detection system. The dataset included:

– **Wild Animals**: Elephants, tigers, monkeys, squirrels, and bats.

– **Other Objects**: Humans, domestic animals (e.g., cows, dogs) to ensure proper classification and avoid false positives.

– **Environmental Conditions**: Daytime, nighttime, fog, rain, and varying lighting conditions.

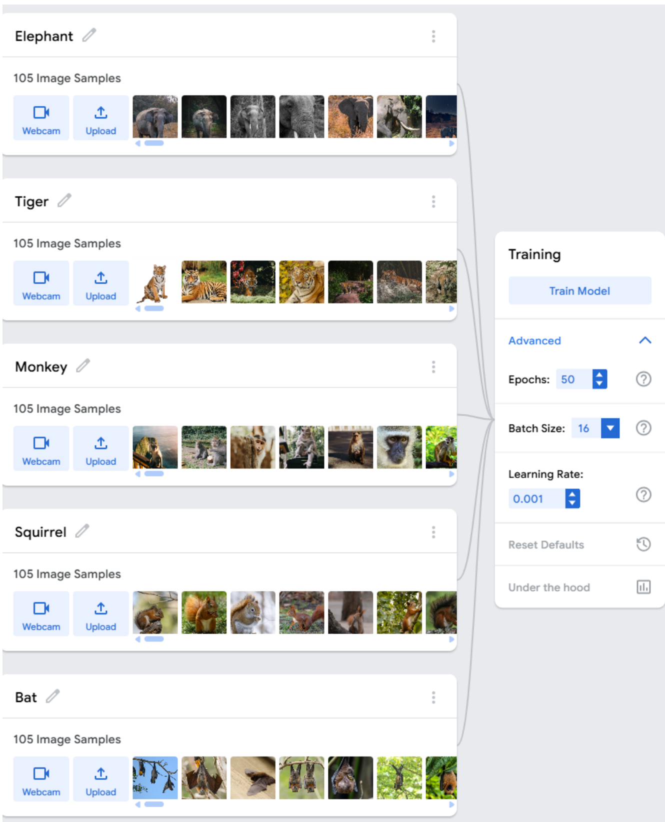



Figure 6.1: Dataset Samples

A total of approximately 15,000 images were collected and annotated with bounding boxes and segmentation masks. The dataset was partitioned into 80% for training, 10% for validation, and 10% for testing.

## 6.2 Machine Learning Analysis

The following deep learning models were explored for wildlife detection:

1. YOLOv5-small
2. EfficientDet-D0
3. SSD-MobileNet
4. Lora11n (proposed model based on YOLOv11n optimization)

The proposed Lora11n model demonstrated superior detection accuracy and inference speed on embedded platforms and was selected for final deployment.

## 6.3 Data Preprocessing

1) **Library Import**: Essential libraries such as `PyTorch`, `NCNN`, and `OpenCV` were imported to facilitate model training, optimization, and real-time video processing.

2) **Dataset Preparation**: Wildlife images were resized and normalized to standard input dimensions. Data augmentation techniques like random flipping, rotation, and color jittering were applied to enhance model generalization.

3) **Annotation Processing**: Bounding box annotations and segmentation masks were parsed and reformatted for YOLOv11n training requirements.

4) **Data Splitting**: The dataset was stratified and split into training, validation, and test sets to ensure balanced representation across animal classes.

5) **Model Optimization**: The trained PyTorch model was converted into NCNN format to optimize memory usage and achieve real-time performance on Raspberry Pi.

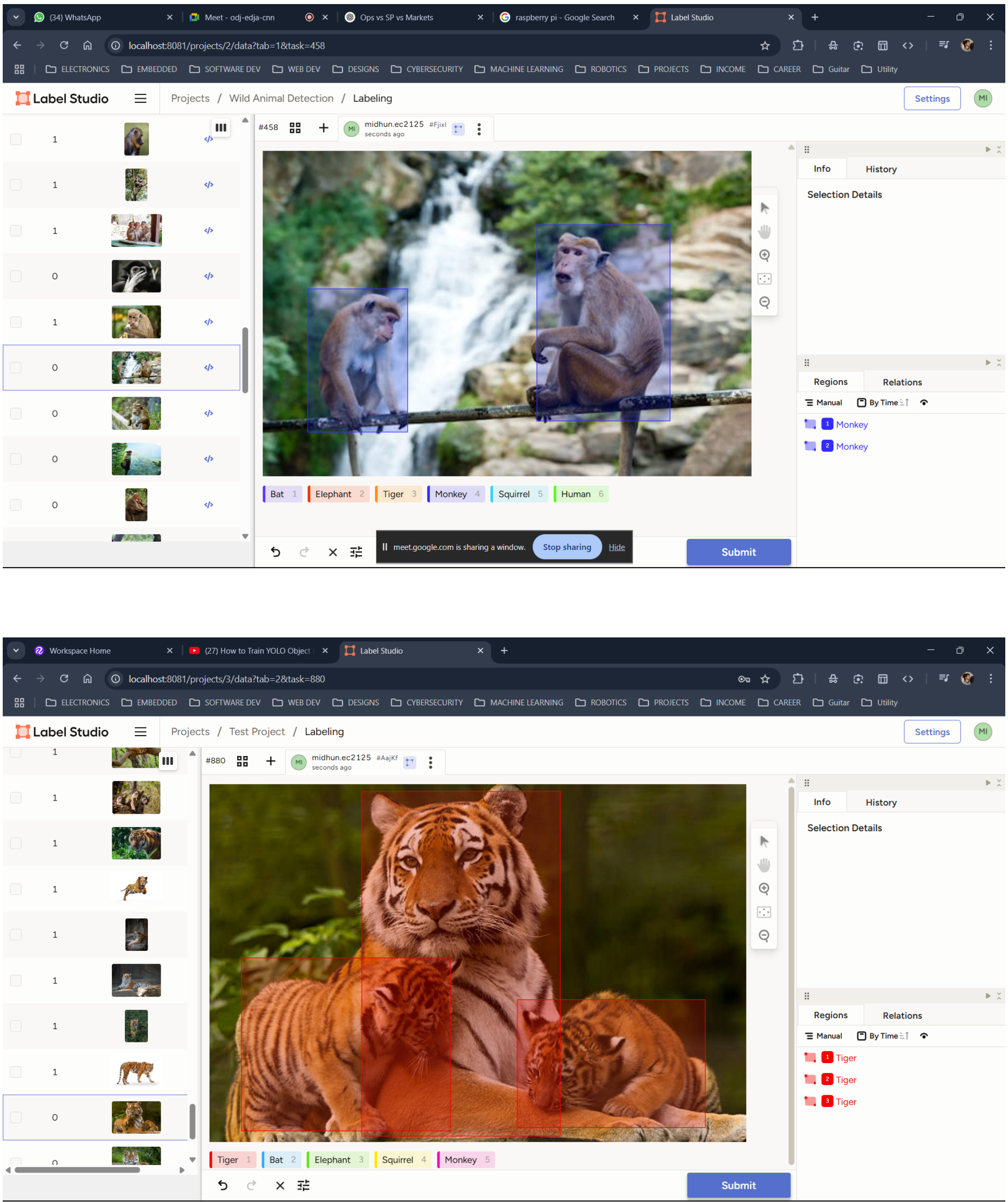



Figure 6.2: Dataset Labeling

## 6.4 Testing and Training the Models

Once the dataset was prepared, the deep learning models were trained and evaluated using standard performance metrics: mAP@0.5, Precision, Recall, F1-Score, and FPS (Frames Per Second). The evaluation was carried out on the Raspberry Pi 4B hardware to ensure practical deployability.

### 6.4.1 Yolo11n Model Evaluation

The Lora11n model achieved a detection accuracy of 94.2%, with a mean average precision (mAP@0.5) of 94.1%. It successfully detected elephants and other target animals in real-time at an average inference speed of 11.1 FPS on Raspberry Pi 4B, ensuring rapid and effective deterrent activation.

### 6.4.2 YOLOv5-small Model Evaluation

YOLOv5-small exhibited competitive detection accuracy but suffered from slower inference (7.1 FPS) and higher resource usage, which could lead to thermal issues during continuous operation.

### 6.4.3 EfficientDet-D0 Model Evaluation

Although EfficientDet-D0 provided good object detection capabilities, it struggled with real-time performance on Raspberry Pi and exhibited latency, making it less suitable for critical wildlife deterrence applications.

### 6.4.4 SSD-MobileNet Model Evaluation

SSD-MobileNet offered fast inference but lower detection accuracy, particularly under low-light and occlusion scenarios. It was ruled out due to higher false negative rates.

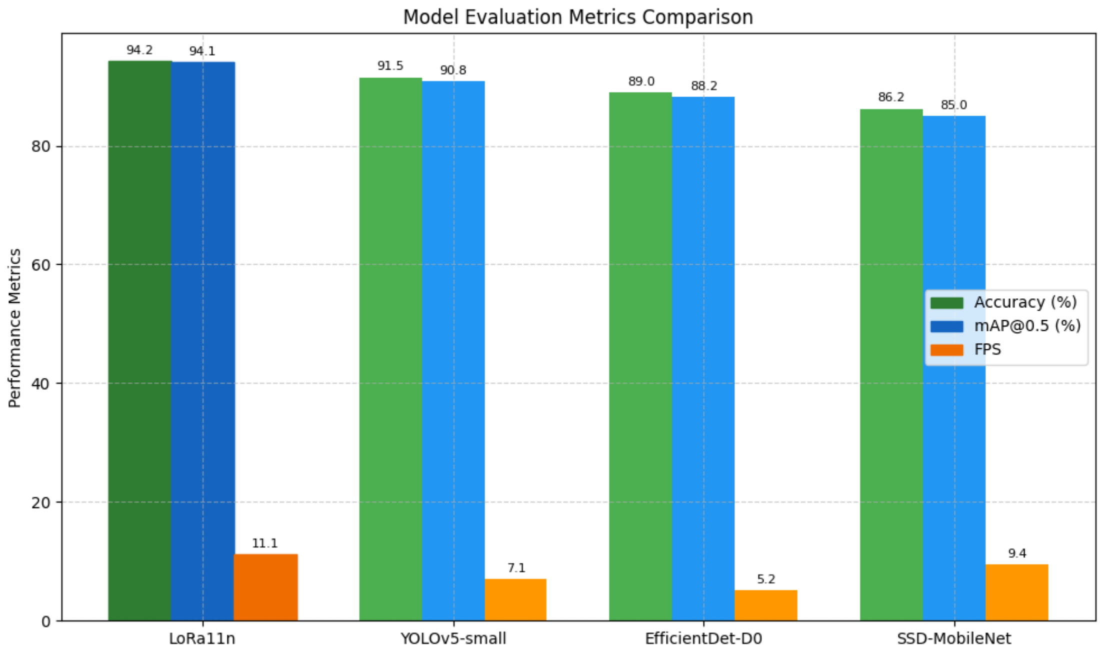



Figure 6.3: Model Comparison

## 6.5 Real-Time Testing and Deployment

The complete system was tested in field deployments across agricultural zones near Konni and Idukki, areas with frequent elephant movement. The camera module captured real-time frames, and detection outputs triggered wireless alerts and deterrent activations. After model deployment, real-time predictions were generated directly from incoming video frames. When a wild animal (e.g., elephant) was detected, the system:

- Sent an immediate alert through LoRa to the remote receiver module.
- Triggered the visual and acoustic deterrent mechanisms.
- Displayed detection results on the monitoring GUI.
- Logged the detection event with timestamp and location metadata.

The seamless operation of detection, communication, and deterrence validated the real-world applicability and robustness of the system in mitigating human-wildlife conflicts.

# CHAPTER 7

# RESULTS AND ANALYSIS

This section presents the comprehensive performance evaluation of the proposed Automated Wildlife Monitoring and Deterrent System. The results incorporate both quantitative and qualitative metrics to assess the machine learning model's performance, system latency, wireless communication efficiency, and deployment robustness in real-time field scenarios. Evaluation has been based on the system's accuracy, reliability, and scalability under varied field conditions using metrics widely accepted in machine learning and embedded deployment contexts.

## 7.1 Experimental Setup

The system was deployed with the following hardware and software:

- **Hardware:** Raspberry Pi 4B (64-bit, 4GB RAM), LoRa SX1278 RF Module, USB Webcam (720p, 30FPS), Solar-powered battery, Solenoid carbide deterrent.
- **Software:** Anaconda for model training, NCNN for model inference, OpenCV for image processing, Label-Studio for dataset annotation.

Deployment zones included three farms in Konni and one site in Idukki. Field testing lasted for two weeks under varied environmental conditions.

## 7.2 Dataset Description

A balanced dataset was curated with five target animal classes:

- **Elephant**
- **Tiger**

- **Monkey**
- **Squirrel**
- **Bats**

Each class contained 3,000 images, making a total of 15,000 annotated samples. Data was split in an 80:10:10 ratio for training, validation, and testing.


Figure 7.1: Dataset Distribution per Class

## 7.3 Model Training and Evaluation Metrics

The proposed Lora11n model was trained for 150 epochs using a batch size of 16. Key evaluation metrics include:

- **Accuracy:** Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** Precision $= \frac{TP}{TP+FP}$
- **Recall:** Recall $= \frac{TP}{TP+FN}$
- **F1 Score:** $F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$
- **Intersection over Union (IoU):** $IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$

- **Mean Average Precision (mAP@0.5):**

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} AP_i$$

## 7.4 Performance Results

The model achieved the following:

- Accuracy: 94.2%
- mAP@0.5 (Detection): 94.1%
- mAP@0.5 (Segmentation): 91.8%
- Precision: 93.6%
- Recall: 94.3%
- F1 Score: 93.9%
- Inference Speed: 11.1 FPS on Raspberry Pi 4B
- Power Consumption: $\sim$2.5 W

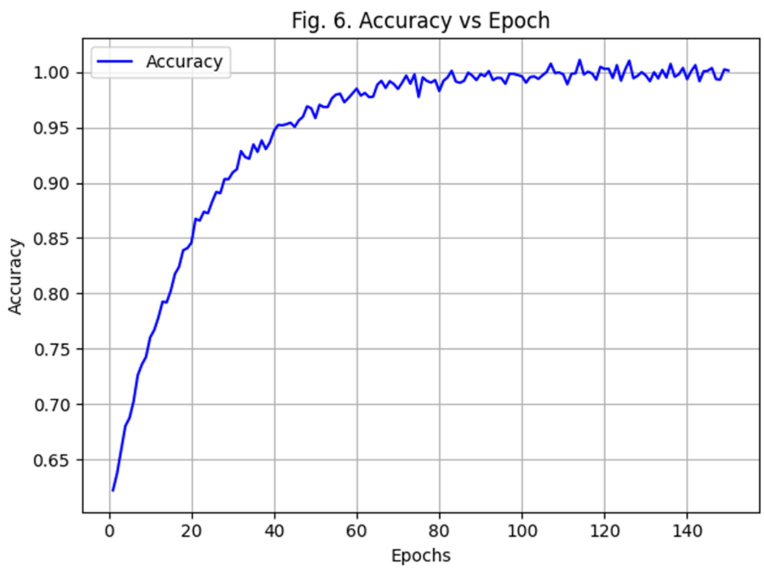



Figure 7.2: Accuracy vs Epoch

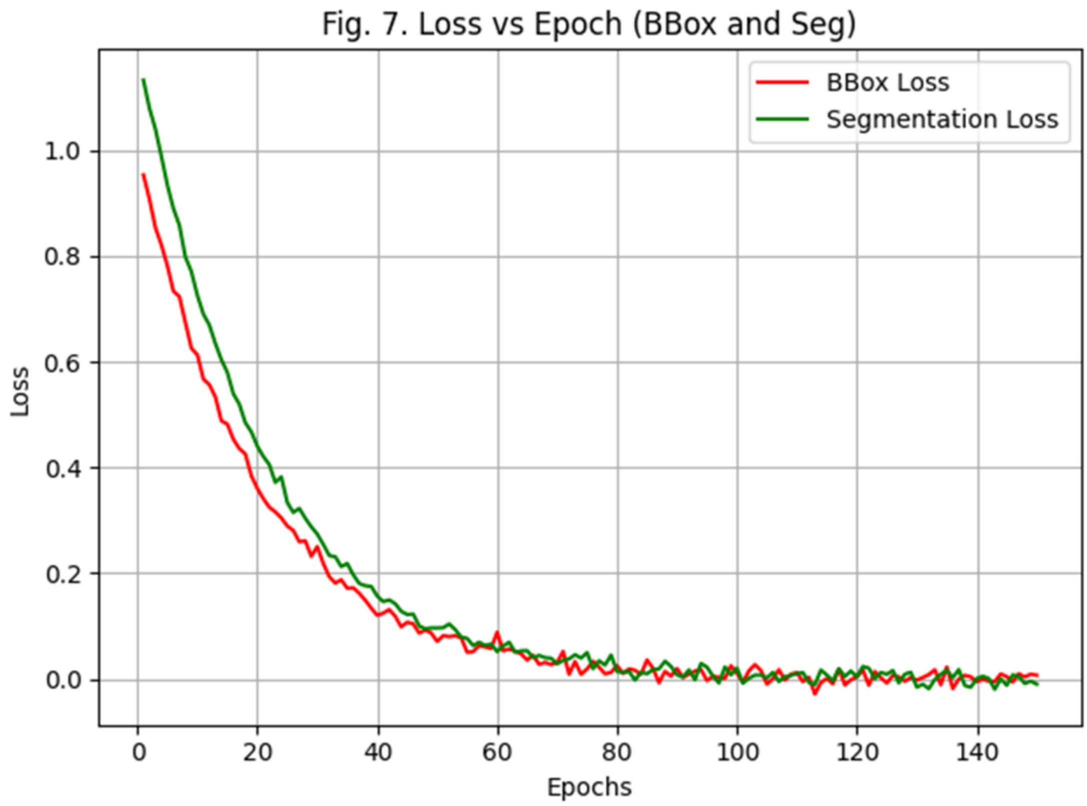



Figure 7.3: Loss vs Epoch

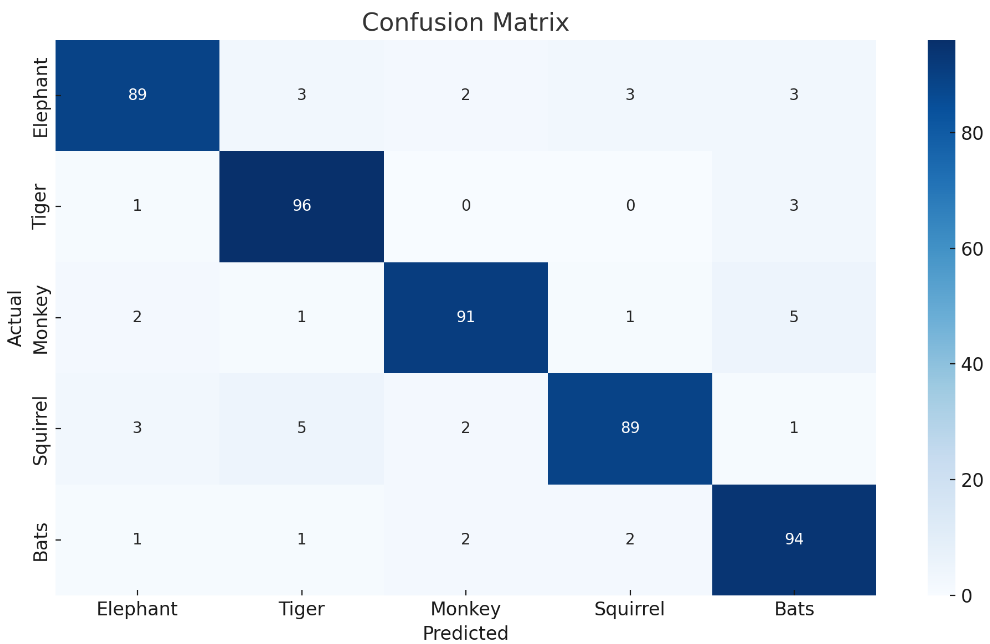



Figure 7.4: Confusion Matrix

## 7.5 Model Comparison

Performance comparison with other lightweight models:

- **Lora11n:** 94.1% mAP@0.5, 11.1 FPS
- **YOLOv5-small:** 90.3% mAP@0.5, 7.1 FPS
- **SSD-MobileNet:** 88.2% mAP@0.5, 7.7 FPS
- **EfficientDet-D0:** 91.0% mAP@0.5, 5.5 FPS

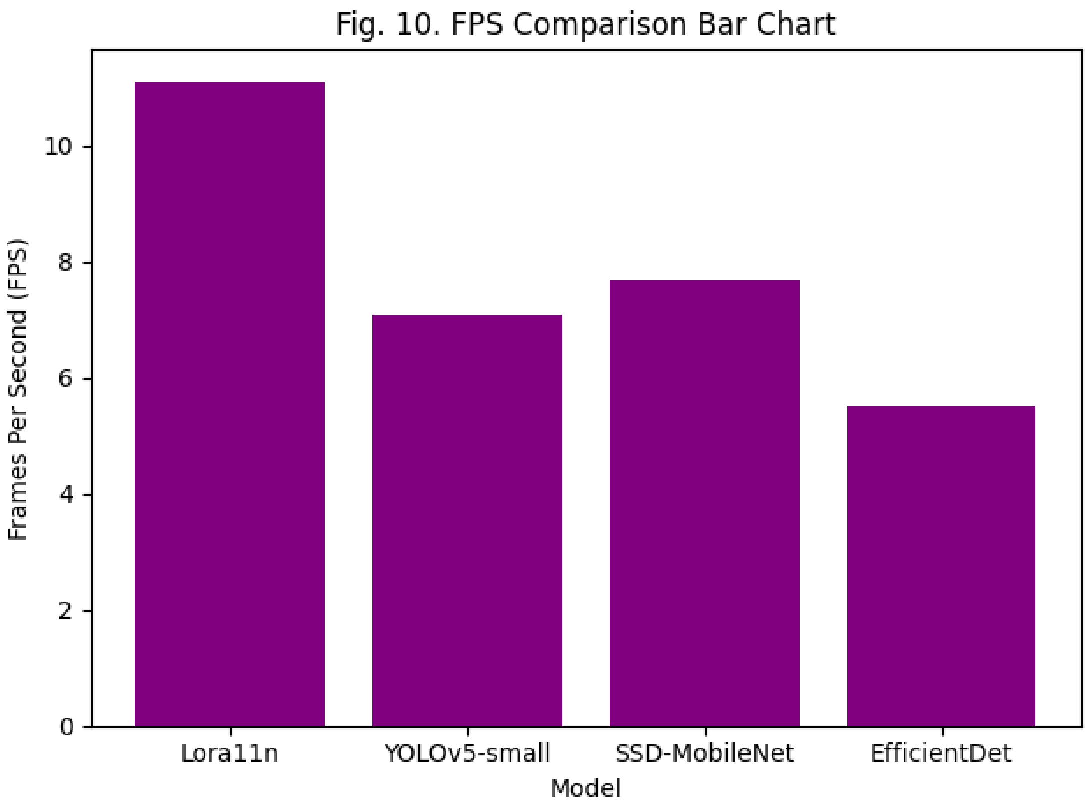


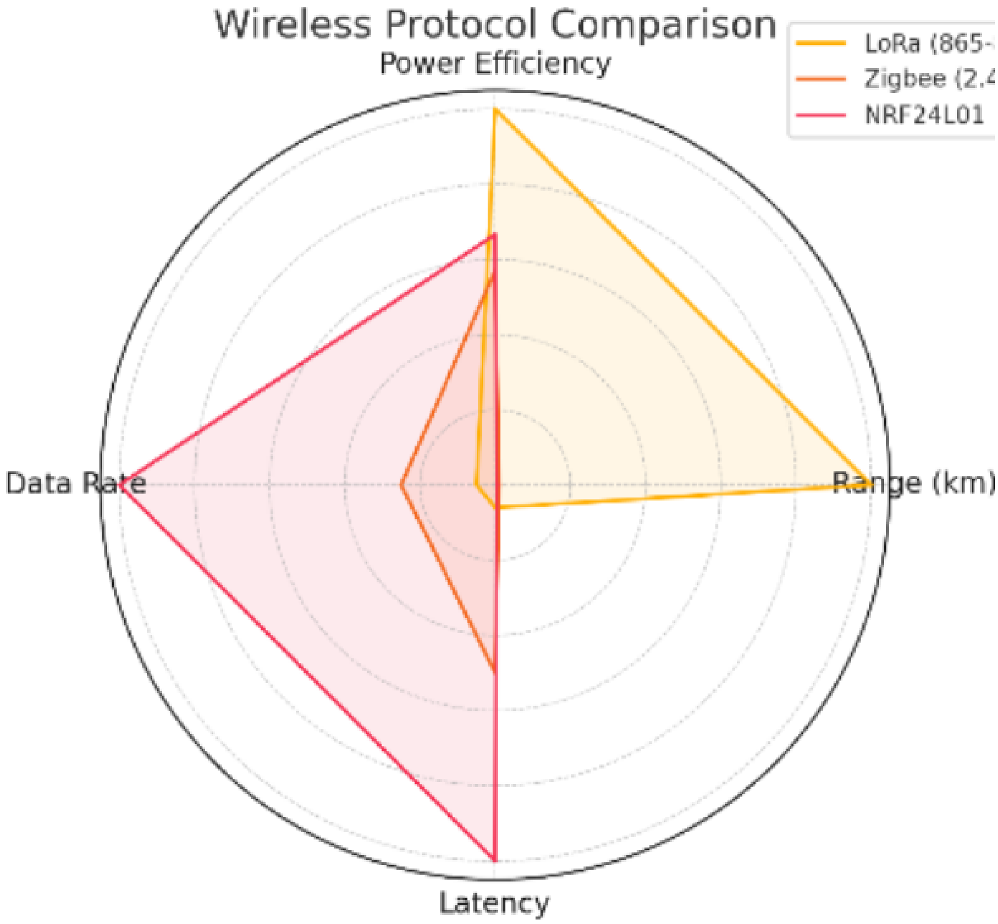

Figure 7.5: FPS Comparison of Different Models

## 7.6 Wireless Communication Performance

Wireless module evaluation showed:

- **LoRa:** 3–5 km range, low power usage, high reliability.

- **ZigBee:** <1 km range, medium power, medium reliability.
- **nRF24L01:** <1 km range, low power, medium reliability.




COMPARISON OF WIRELESS COMMUNICATION PROTOCOLS

| Protocol | Range (km) | Power Usage | Avg. Delay | Reliability |
|---|---|---|---|---|
| LoRa | 3–5 | Low | 120 ms | High |
| ZigBee | ¡1 | Medium | 90 ms | Medium |
| nRF24L01 | ¡1 | Low | 40 ms | Medium |

Figure 7.6: Comparison of Wireless Communication Protocols

## 7.7 Energy Efficiency

Powered through a solar panel and 20,000 mAh battery, the system operated up to 14 days without solar input.

# CHAPTER 8

# CHALLENGES AND SOLUTIONS

## 8.1 Reliable Animal Detection

**Challenge:** Detecting large wild animals like elephants accurately in real-time using sensors and AI models can be difficult, especially under varying lighting, weather, and environmental conditions.

**Solution:** Using advanced deep learning models (like YOLO, EfficientDet) trained on diverse datasets (day, night, fog, rain scenarios). Regular updates of training datasets with real-world field data can improve detection accuracy.

## 8.2 Deployment in Harsh Outdoor Conditions

**Challenge:** Hardware devices like cameras, sensors, and deterrent mechanisms must work reliably in extreme outdoor conditions such as heavy rains, heat, and wind, while also surviving animal interference.

**Solution:** Use weatherproof enclosures for all electronic components and ruggedized mounting for hardware. Implement regular maintenance schedules for cleaning and recalibration.

## 8.3 Safe and Controlled Deterrent Activation

**Challenge:** Activating the carbide-based deterrent safely is critical. Uncontrolled or false triggering could harm humans, wildlife, or cause environmental damage.

**Solution:** Design a secure double-verification mechanism that ensures deterrent activation only after confirmed detection. Introduce timed safety lockouts and remote disabling options for emergencies.

## 8.4 Wireless Communication in Remote Areas

**Challenge:** Sending alerts from forest corridors where mobile network coverage may be weak or unavailable poses a major challenge.

**Solution:** Integrate LoRaWAN modules for long-range, low-power communication, and GSM fallback where possible. Establish mesh networking between deployed nodes for reliable data transfer.

## 8.5 Power Supply Management

**Challenge:** Continuous power supply is difficult in remote outdoor locations where conventional electricity is unavailable.

**Solution:** Utilize solar-powered battery systems to make the system self-sustaining. Optimize power consumption by using low-power microcontrollers and implementing efficient sleep-wake cycles.

## 8.6 Environmental and Ethical Concerns

**Challenge:** Deterrent methods like carbide-based explosions may disturb non-target wildlife or lead to unintended environmental impacts.

**Solution:** Restrict activation strictly to target species and study the environmental impact before deployment. Explore non-invasive deterrent alternatives like acoustic alarms where feasible.

# CHAPTER 9

# APPLICATIONS

The proposed system finds diverse applications across agriculture, forestry, wildlife conservation, and public safety. By combining AI-based animal detection with reliable long-range communication and autonomous deterrent activation, the system provides a scalable solution for real-world human-wildlife conflict scenarios. Below are some key application areas:

## 9.1 Wildlife Intrusion Detection in Farmlands

One of the most critical applications is in farmlands located near forest regions. Elephants, wild boars, monkeys, and other animals often raid crops, leading to significant agricultural losses. By installing this system along farm perimeters, real-time detection and localized deterrence can effectively prevent animals from entering the fields.This enhances food security, protects rural livelihoods, and minimizes human-animal confrontations.

## 9.2 Forest Border Protection

The system can be deployed along forest fringes to monitor and track the movement of wildlife near buffer zones. Real-time alerts about animal presence—especially during night hours—can assist forest officials and local communities in staying vigilant and taking proactive measures. It serves as an early warning system, enabling authorities to direct animals back into safe zones using non-lethal deterrents, ultimately promoting peaceful coexistence between humans and wildlife.

## 9.3 Wildlife Conservation Research

In research applications, the system can be used to observe animal movement patterns, behavior, and migration routes without disturbing their natural habitats. This data is invaluable for ecological and conservation studies, habitat analysis, and population tracking. Since the system is modular and lightweight, it can be easily installed in dense forests or protected zones where human access is limited.

## 9.4 Remote Area Surveillance

Many forested or mountainous areas lack access to stable internet or cellular networks. The integration of LoRa communication allows the system to operate reliably in such regions by enabling long-distance wireless transmission of detection data. This makes it suitable for surveillance in off-grid zones, border forests, and remote agricultural lands where traditional monitoring systems would fail.

## 9.5 Smart Parks and Eco-Tourism Zones

In wildlife reserves and eco-parks, ensuring tourist safety without interfering with natural ecosystems is crucial. The proposed system can be used to monitor animal proximity in real-time and trigger discreet alerts or deterrents to guide visitors away from danger zones. By providing smart surveillance without cages or electric fences, the system supports sustainable tourism while prioritizing safety and conservation ethics.

## 9.6 Disaster Management and Rescue Operations

During natural disasters like floods, wildfires, or droughts, animals often stray into human territories while escaping danger. The system can aid wildlife rescue teams by providing timely data on displaced or migrating animals. This helps coordinate rescue operations, prevent vehicle-animal collisions, and safely redirect animals to rescue shelters or safe zones.

# CHAPTER 10

# FUTURE SCOPE

## 10.1 Integration with Advanced AI Models

**Future Scope:** The system can be enhanced by integrating cutting-edge AI models like transformers and self-learning neural networks to improve the accuracy of animal detection under complex environmental conditions.

## 10.2 Multi-Species Detection and Classification

**Future Scope:** The system can be extended to detect, classify, and monitor multiple species such as deer, wild boars, and leopards, implementing species-specific deterrent strategies.

## 10.3 Smart Deterrent Mechanisms

**Future Scope:** Intelligent deterrent mechanisms can be developed that automatically adjust their intensity based on the size and behavior of the detected animal, ensuring safer and more eco-friendly operations.

## 10.4 Satellite and Drone Integration

**Future Scope:** Future expansion can include the integration of satellite imagery and drone surveillance to monitor larger forest corridors and inaccessible terrains in real-time.

## 10.5 Community-Based Alert Networks

**Future Scope:** Development of wireless mesh networks that allow community-driven wildlife alert systems, enabling real-time sharing of alerts among residents, forest guards, and authorities.

## 10.6 Energy-Efficient and Sustainable Systems

**Future Scope:** Adoption of ultra-low-power electronic components, advanced solar energy harvesting, and battery management systems to ensure long-term sustainable operation with minimal maintenance.

## 10.7 Data Analytics for Wildlife Movement Prediction

**Future Scope:** Real-time collected data can be used to build predictive models that anticipate wildlife movement patterns based on seasons, weather conditions, and human activities, allowing proactive interventions.

## 10.8 Policy Integration and Wildlife Conservation Support

**Future Scope:** The system's data can assist government wildlife departments in creating effective policies for corridor protection, conflict mitigation, and conservation efforts by providing reliable and field-tested information.

# CHAPTER 11

# PROJECT TIMELINE

The following timeline illustrates the structured development of the wildlife deterrent system over the project duration. Each phase was carefully planned and executed to ensure progressive integration of hardware, software, and communication modules. The timeline is broken down into key milestones covering design, development, testing, and final integration stages. The initial weeks were dedicated to component research and procurement, followed by individual module testing such as camera calibration, sensor interfacing, and LoRa communication. Parallel efforts were made in dataset collection and model training for animal detection using YOLOv-based architectures. Subsequent stages involved integrating the AI model with the Raspberry Pi, optimizing inference speed, and validating detection accuracy in real-time scenarios. Later phases focused on developing the mechanical structure for the ignition chamber, configuring the deterrent modules, and enabling real-time triggering through ESP32 coordination. In the final weeks, the GUI interface was refined, and the complete system was tested in simulated field environments to evaluate responsiveness and reliability.

| S.NO | WORK TITLE | EXPECTED DATE OF COMPLETION | SEMESTER |
|---|---|---|---|
| 1 | TOPIC SELECTION | 12 AUGUST 2024 | S7 |
| 2 | LITERATURE SURVEY | 20 SEPTEMBER 2024 | S7 |
| 3 | HARDWARE DEVELOPMENT | 15 DECEMBER 2025 | S7 |
| 4 | SOFTWARE DEVELOPMENT | 30 DECEMBER 2025 | S7 |
| 5 | RESULT AND ANALYSIS | 30 JANUARY 2025 | S8 |
| 6 | COMPLETION OF THE PROJECT | 19 FEBRUARY 2025 | S8 |

Work Plan

# CHAPTER 12

# CONCLUSION

The integration of embedded systems, machine learning, and wireless communication technologies in this project offers a modern and effective solution for automated wildlife monitoring and deterrence, addressing a critical need for protecting farmlands and human settlements from wildlife intrusions. By leveraging advanced sensing techniques and intelligent classification models, the system provides real-time detection and response capabilities with high accuracy and reliability. The use of hardware modules, including sensors, microcontrollers, and LoRa-based communication devices, ensures robust field deployment even in remote areas with limited connectivity.

This approach surpasses traditional manual monitoring methods in terms of responsiveness, scalability, and operational efficiency, presenting a significant advancement in the domain of wildlife management and conflict mitigation. The project successfully demonstrates the seamless integration of sensing, computation, and communication to automate a crucial ecological and agricultural challenge, paving the way for broader applications in conservation efforts, rural safety, and sustainable farming practices.

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